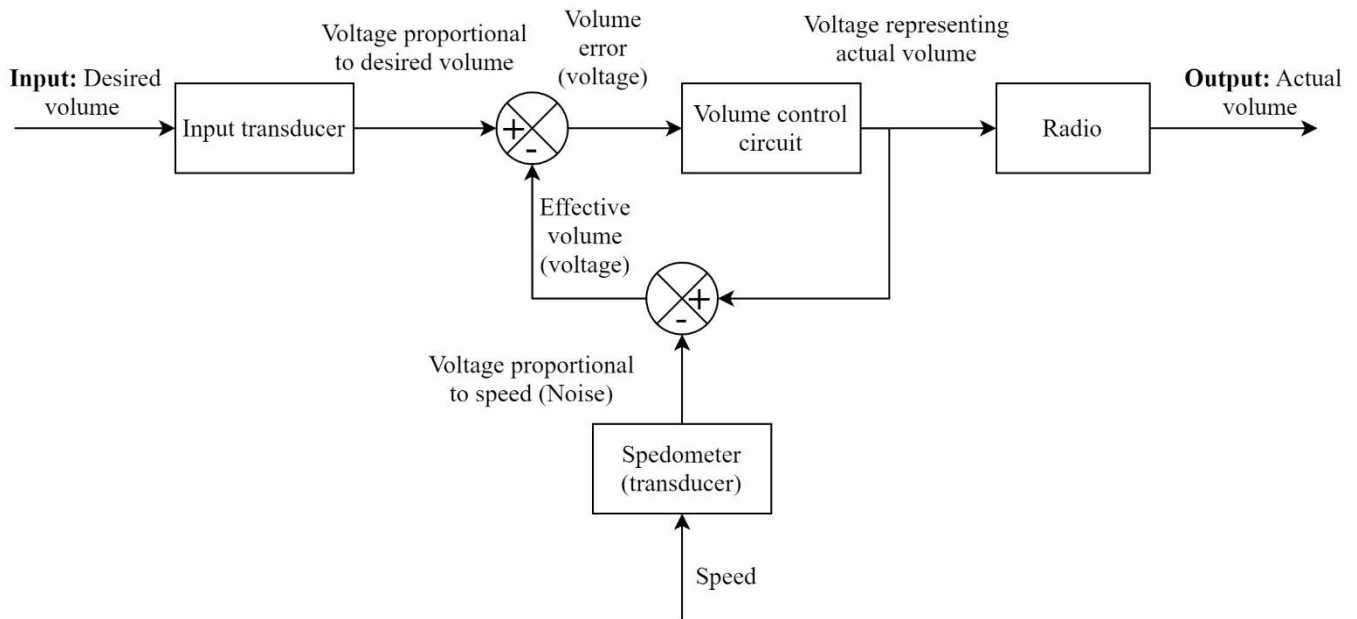
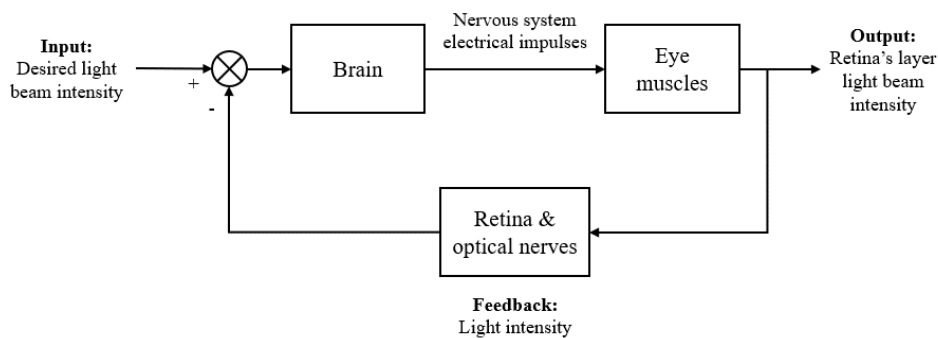


Chapter 1 Homework

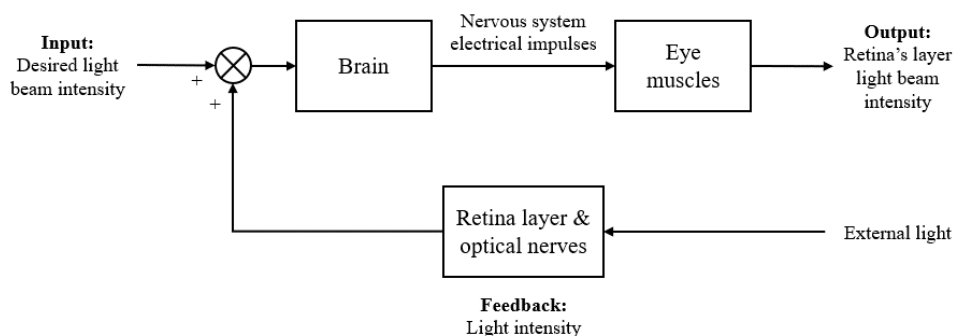
Question 3



Question 6(a)



Question 6(b)



When the incident light is smaller than the diameter of the pupil, the feedback loop is broken and the external light acts as the feedback.

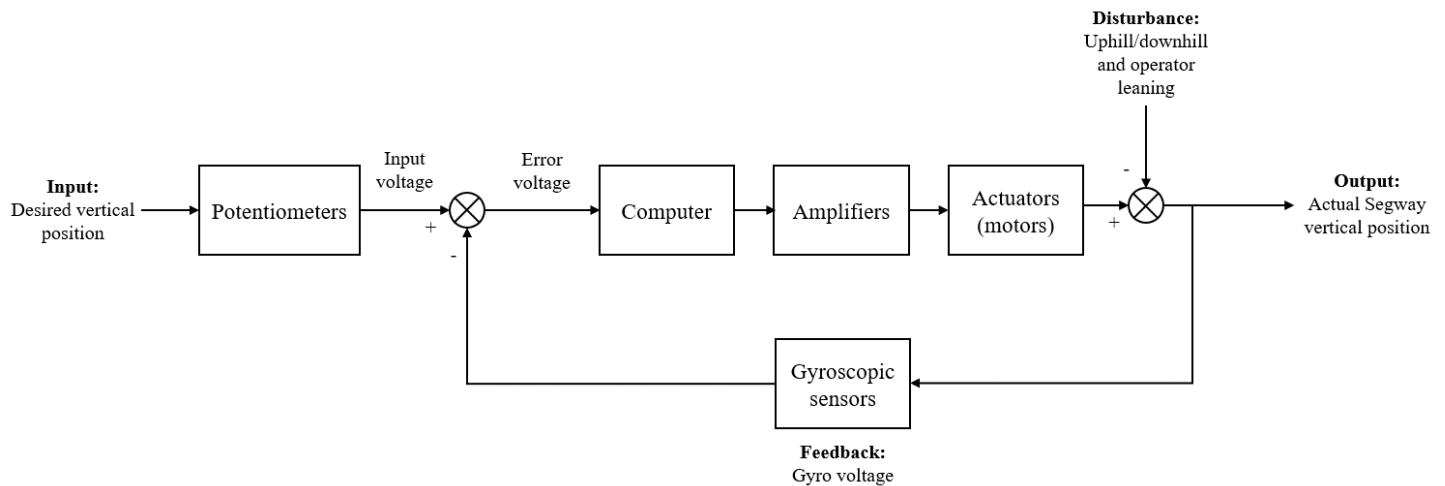
If the narrow beam of light varies in intensity, in this case, a sinusoidal function, the pupil's diameter will also vary sinusoidally but with a delay.

Question 6(c)

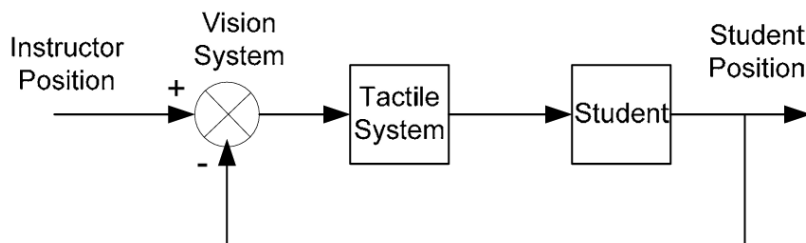
If there is no delay, the pupil would begin to contract instantly to the point where a small amount of light goes in, at that point it would stop contracting and the pupil would remain with a fixed diameter.

With the delay present, it would take 300 milliseconds for the pupil to respond to the change in the incident light. The pupil would similarly contract to the point where a small amount of light goes in and then remain at a fixed diameter.

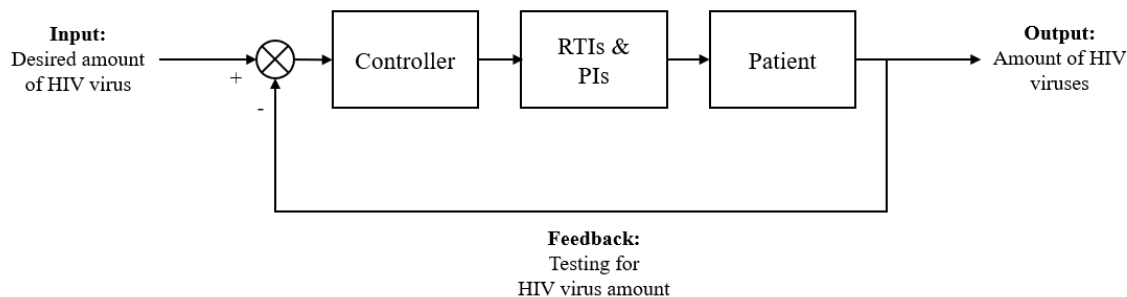
Question 7



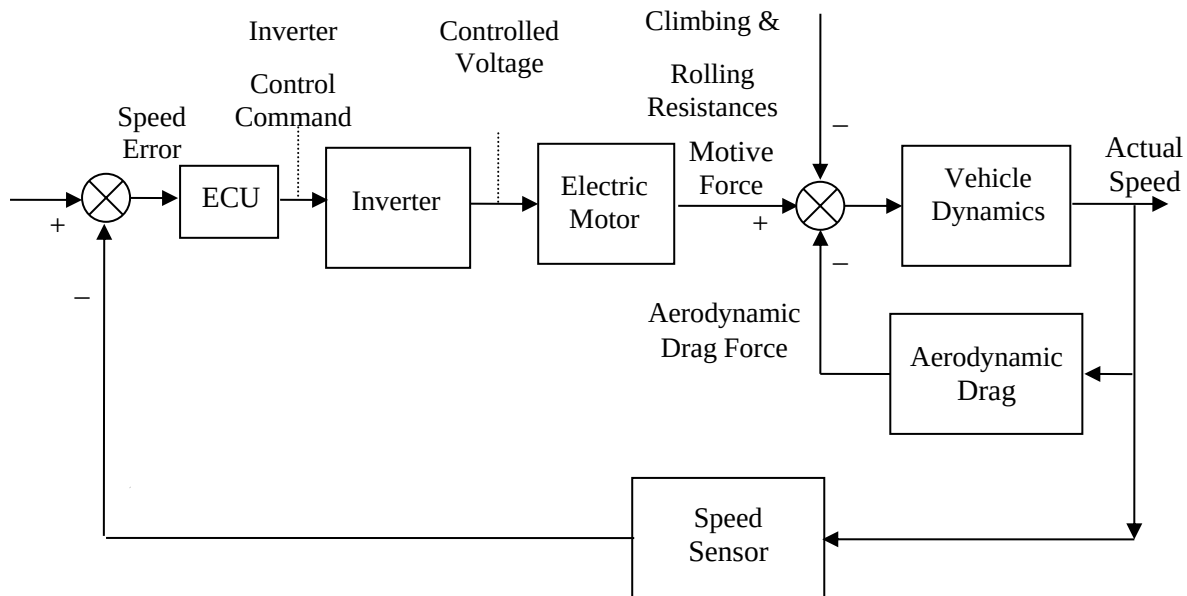
Question 9



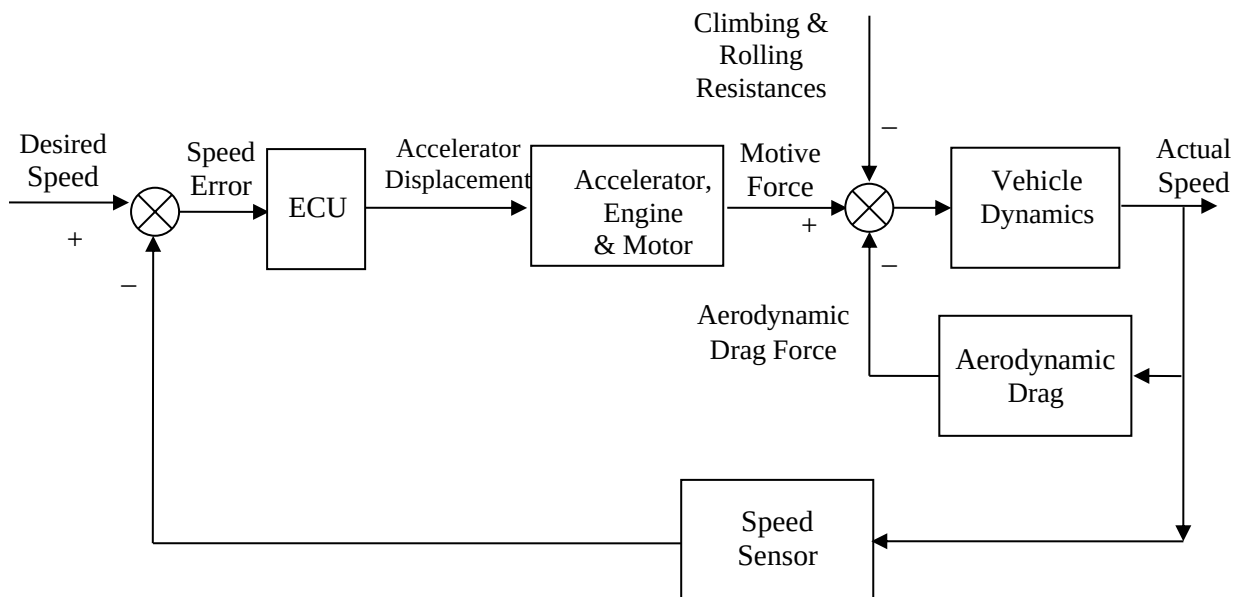
Question 17



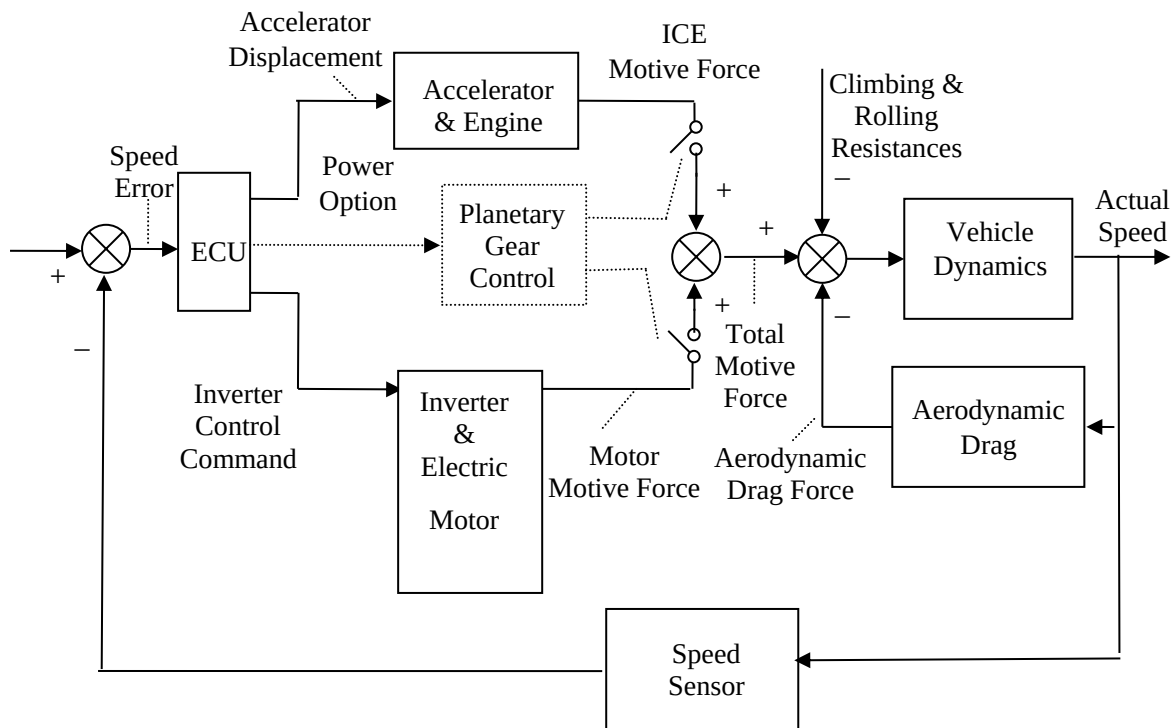
Question 18(a)



Question 18(b)



Question 18(c)



Chapter 2A Homework

Question 7

Assume zero initial conditions, the Laplace transform is:

$$(s^3 + 3s^2 + 5s + 1)Y(s) = (s^3 + 4s^2 + 6s + 8)X(s)$$

Solve for $Y(s)/X(s)$:

$$\frac{Y(s)}{X(s)} = \frac{s^3 + 4s^2 + 6s + 8}{s^3 + 3s^2 + 5s + 1}$$

Question 9

The transfer function $C(s)/R(s)$ is:

$$\frac{C(s)}{R(s)} = \frac{s^5 + 2s^4 + 4s^3 + s^2 + 4}{s^6 + 7s^5 + 3s^4 + 2s^3 + s^2 + 5}$$

Cross-multiply the transfer function:

$$(s^6 + 7s^5 + 3s^4 + 2s^3 + s^2 + 5)C(s) = (s^5 + 2s^4 + 4s^3 + s^2 + 4)R(s)$$

Assume zero initial conditions, the inverse Laplace transform is:

$$\frac{d^6 c}{dt^6} + 7 \frac{d^5 c}{dt^5} + 3 \frac{d^4 c}{dt^4} + 2 \frac{d^3 c}{dt^3} + \frac{d^2 c}{dt^2} + 5c = \frac{d^5 r}{dt^5} + 2 \frac{d^4 r}{dt^4} + 4 \frac{d^3 r}{dt^3} + \frac{d^2 r}{dt^2} + 4r$$

Question 14(a)

The circuit elements are converted into their Laplace transform equivalents. The equivalent parallel of the rightmost inductor in parallel with the resistor is $Z = s || 2 = \frac{2s}{s+2}$. Applying the voltage divider rule one gets

$$\frac{V_o}{V_i} = \frac{\frac{2s}{s+2}}{2 + \frac{2s}{s+2}} = \frac{s}{2(s+1)}$$

Question 14(b)

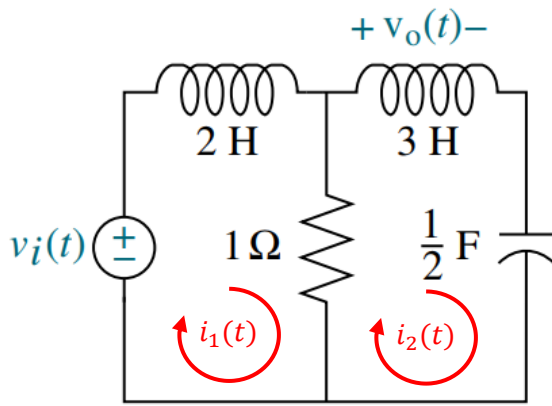
The circuit elements are converted into their Laplace transform equivalents. The rightmost resistor in parallel with the branch with an inductor and a capacitor in series is

$$Z = 1 || \left(2s + \frac{1}{s} \right) = \frac{2s + \frac{1}{s}}{1 + 2s + \frac{1}{s}} = \frac{2s^2 + 1}{2s^2 + s + 1}$$

Applying the voltage divider rule

$$\frac{V_o}{V_i} = \frac{\frac{2s^2 + 1}{2s^2 + s + 1}}{1 + \frac{2s^2 + 1}{2s^2 + s + 1}} = \frac{\frac{1}{4}}{s^2 + \frac{s}{4} + \frac{1}{2}}$$

Question 15(a)



The mesh equations are:

$$(2s + 1)I_1(s) - I_2(s) = V_i(s) \dots [1]$$

$$-I_1(s) + \left(3s + 1 + \frac{2}{s}\right)I_2(s) = 0 \dots [2]$$

Solve [2] for $I_1(s)$:

$$I_1(s) = \left(3s + 1 + \frac{2}{s}\right)I_2(s) \dots [2]$$

Substitute [2] into [1]:

$$(2s + 1)\left(3s + 1 + \frac{2}{s}\right)I_2(s) - I_2(s) = V_i(s)$$

$$\frac{I_2(s)}{V_i(s)} = \frac{s}{6s^3 + 5s^2 + 4s + 2} \dots [3]$$

The output voltage is the voltage across the inductor:

$$v_L(t) = v_o(t) = L \frac{di_2(t)}{dt}$$

$$v_o(t) = 3 \frac{di_2(t)}{dt}$$

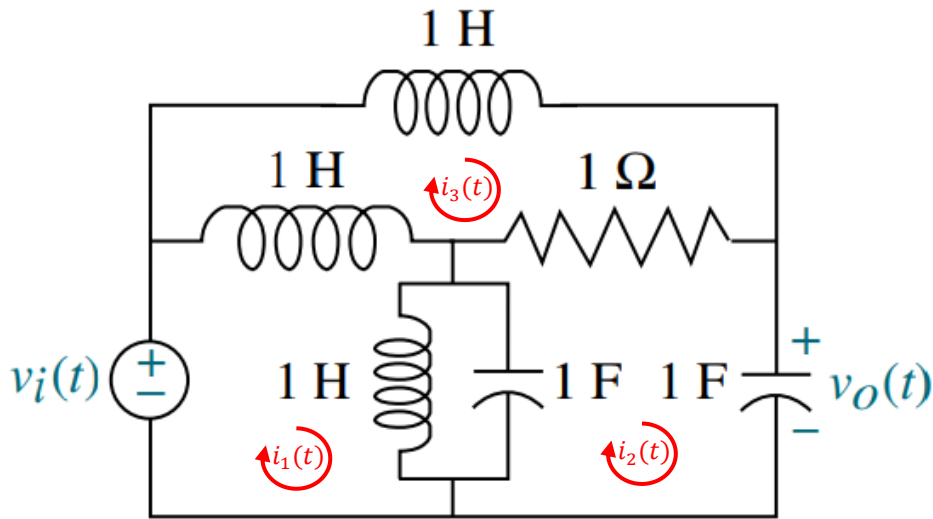
$$V_o(s) = 3sI_2(s)$$

$$I_2(s) = \frac{V_o(s)}{3s} \dots [4]$$

Substitute [4] into [3]:

$$\frac{V_o(s)}{V_i(s)} = \frac{3s^2}{6s^3 + 5s^2 + 4s + 2}$$

Question 15(b)



The equivalent impedance of the parallel inductor and capacitor:

$$\frac{s(1/s)}{s + 1/s} = \frac{s}{s^2 + 1}$$

The mesh equations are:

$$\left(s + \frac{s}{s^2 + 1}\right) I_1(s) - \left(\frac{s}{s^2 + 1}\right) I_2(s) - sI_3(s) = V_i(s) \dots [1]$$

$$-\left(\frac{s}{s^2 + 1}\right) I_1(s) + \left(\frac{s}{s^2 + 1} + 1 + \frac{1}{s}\right) I_2(s) - I_3(s) = 0 \dots [2]$$

$$-sI_1(s) - I_2(s) + (2s + 1)I_3(s) = 0 \dots [3]$$

Solve for $I_2(s)$ from the mesh equations:

$$I_2(s) = \left[\frac{s(s^2 + 2s + 2)}{s^4 + 2s^3 + 3s^2 + 3s + 2} \right] V_i(s)$$

$$\frac{I_2(s)}{V_i(s)} = \frac{s(s^2 + 2s + 2)}{s^4 + 2s^3 + 3s^2 + 3s + 2} \dots [4]$$

The output voltage is the voltage across the capacitor:

$$v_o(t) = v_c(t) = \frac{1}{C} \int i_2(t) dt$$

$$v_o(t) = \int i_2(t) dt$$

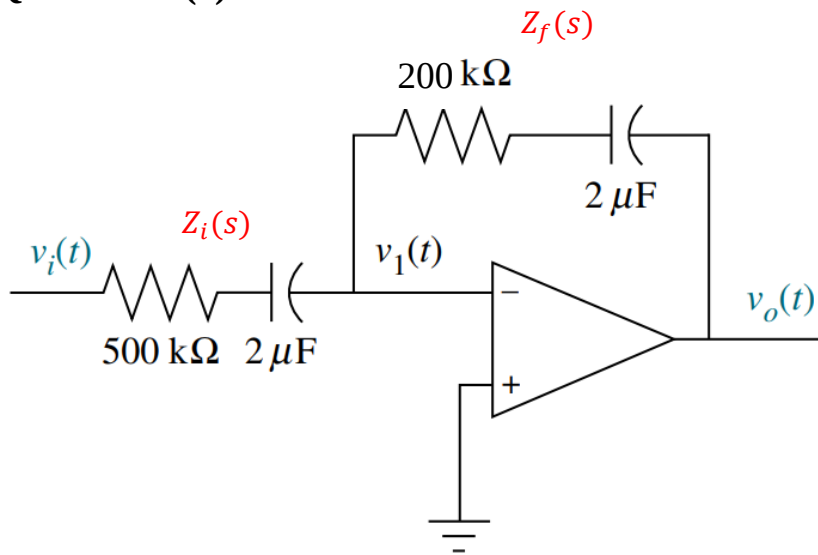
$$V_o(s) = \frac{1}{s} I_2(s)$$

$$I_2(s) = sV_o(s) \dots [5]$$

Substitute [5] into [4]:

$$\frac{V_o(s)}{V_i(s)} = \frac{s^2 + 2s + 2}{s^4 + 2s^3 + 3s^2 + 3s + 2}$$

Question 17(a)



$Z_i(s)$ is the Laplace equivalent series impedance of the input $500\text{ k}\Omega$ and $2\text{ }\mu\text{F}$ components:

$$Z_i(s) = 500 \times 10^3 + \frac{1}{2 \times 10^{-6}s}$$

$$Z_i(s) = \frac{5 \times 10^5}{s} + 5 \times 10^5$$

$$Z_i(s) = 1 \times 10^5 \left(\frac{5}{s} + 5 \right)$$

$Z_f(s)$ is the Laplace equivalent series impedance of the feedback $200\text{ k}\Omega$ and $2\text{ }\mu\text{F}$ components:

$$Z_f(s) = 200 \times 10^3 + \frac{1}{2 \times 10^{-6}s}$$

$$Z_f(s) = \frac{5 \times 10^5}{s} + 2 \times 10^5$$

$$Z_f(s) = 1 \times 10^5 \left(\frac{5}{s} + 2 \right)$$

Inverting op-amp configuration, substitute and simplify:

$$V_o(s) = -\frac{Z_f(s)}{Z_i(s)} V_i(s)$$

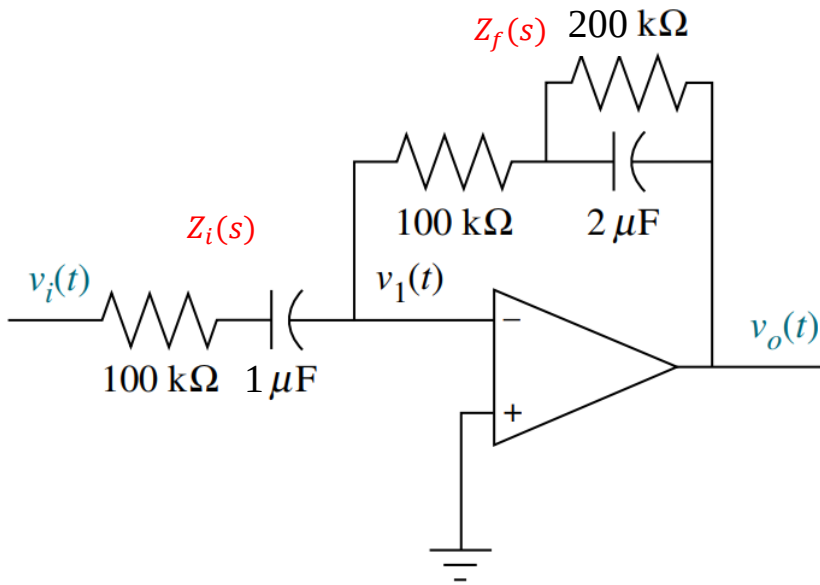
$$\frac{V_o(s)}{V_i(s)} = -\frac{1 \times 10^5 \left(\frac{5}{s} + 2 \right)}{1 \times 10^5 \left(\frac{5}{s} + 5 \right)}$$

$$\frac{V_o(s)}{V_i(s)} = -\frac{\left(\frac{5}{s} + 2 \right)}{\left(\frac{5}{s} + 5 \right)} \times \frac{(5/s)}{(5/s)}$$

$$\frac{V_o(s)}{V_i(s)} = -\frac{\left(1 + \frac{2s}{5} \right)}{(1 + s)}$$

$$\frac{V_o(s)}{V_i(s)} = -\frac{1}{5} \frac{(2s + 5)}{(s + 1)}$$

Question 17(b)



$Z_i(s)$ is the Laplace equivalent series impedance of the input components:

$$Z_i(s) = 100 \times 10^3 + \frac{1}{1 \times 10^{-6}s}$$

$$Z_i(s) = 1 \times 10^5 \left(\frac{10}{s} + 1 \right)$$

$Z_f(s)$ is the Laplace equivalent series impedance of the feedback components:

$$Z_f(s) = 100 \times 10^3 + \frac{(200 \times 10^3) \left(\frac{1}{2 \times 10^{-6}s} \right)}{200 \times 10^3 + \frac{1}{2 \times 10^{-6}s}}$$

$$Z_f(s) = \frac{1 \times 10^{11}}{s \left(\frac{5 \times 10^5}{s} + 2 \times 10^5 \right)} + 1 \times 10^5$$

$$Z_f(s) = 1 \times 10^5 \left(\frac{10}{2s + 5} + 1 \right)$$

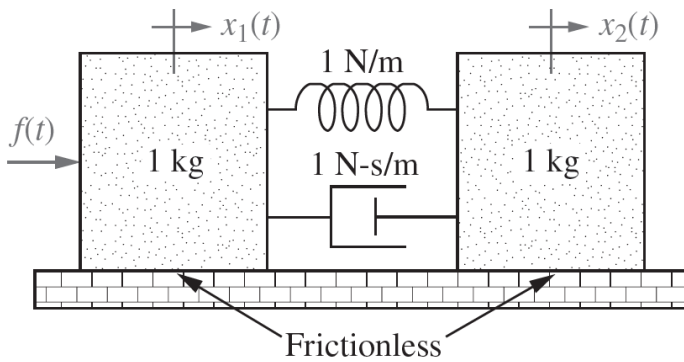
Inverting op-amp configuration, substitute and simplify:

$$V_o(s) = -\frac{Z_f(s)}{Z_i(s)} V_i(s)$$

$$\frac{V_o(s)}{V_i(s)} = -\frac{1 \times 10^5 \left(\frac{10}{2s + 5} + 1 \right)}{1 \times 10^5 \left(\frac{10}{s} + 1 \right)}$$

$$\frac{V_o(s)}{V_i(s)} = -\frac{\left(\frac{10 + (2s + 5)}{2s + 5} \right)}{\frac{1}{s}(10 + s)} = -\frac{s(2s + 15)}{(2s + 5)(s + 10)}$$

Question 20



Writing the equations of motion adding up all forces acting on the block from the spring, damper and external forces. Remember that the spring and damper are common to both blocks and that the mass (inertia s^2 term) of the block is only relevant to that particular block.

For block 1,

$$(s^2 + s + 1)X_1(s) - (s + 1)X_2(s) = F(s)$$

and block 2,

$$-(s + 1)X_1(s) + (s^2 + s + 1)X_2(s) = 0$$

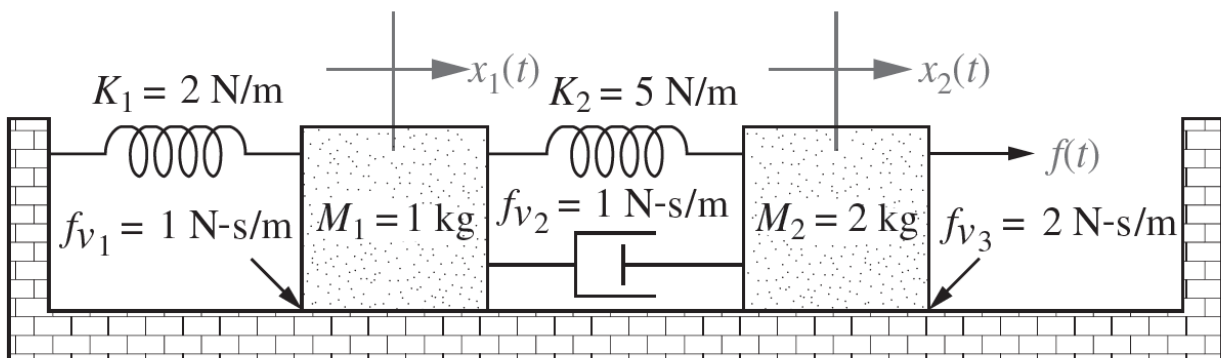
Solving for $X_2(s)$,

$$X_2(s) = \frac{\begin{vmatrix} (s^2 + s + 1) & F(s) \\ -(s + 1) & 0 \end{vmatrix}}{\begin{vmatrix} (s^2 + s + 1) & -(s + 1) \\ -(s + 1) & (s^2 + s + 1) \end{vmatrix}} = \frac{(s + 1)F(s)}{s^2(s^2 + 2s + 2)}$$

From which,

$$\frac{X_2(s)}{F(s)} = \frac{(s + 1)}{s^2(s^2 + 2s + 2)}$$

Question 21



The system has two independent translational displacements, so we can write the following two equations:

$$\begin{aligned} X_1: \quad & (s^2 + 2s + 7)X_1(s) - (s + 5)X_2(s) = 0 \\ X_2: \quad & -(s + 5)X_1(s) + (2s^2 + 3s + 5)X_2(s) = F(s) \end{aligned}$$

Solving we get:

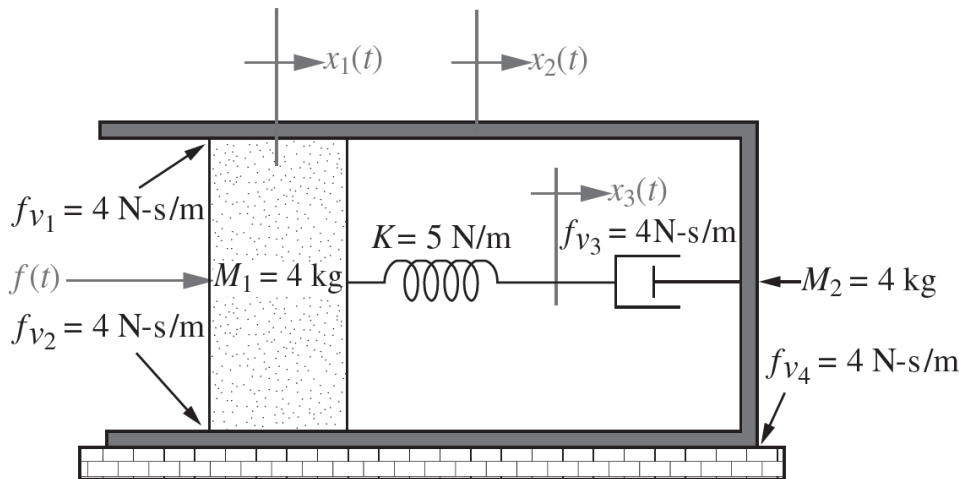
$$X_2(s) = \frac{\begin{vmatrix} s^2 + 2s + 7 & 0 \\ -(s+5) & F(s) \end{vmatrix}}{\begin{vmatrix} s^2 + 2s + 7 & -(s+5) \\ -(s+5) & 2s^2 + 3s + 5 \end{vmatrix}} = \frac{(s^2 + 2s + 7)F(s)}{(s^2 + 2s + 7)(2s^2 + 3s + 5) - (s+5)^2}$$

$$= \frac{(s^2 + 2s + 7)F(s)}{2s^4 + 7s^3 + 24s^2 + 21s + 10}$$

The resulting transfer function can be written as

$$\frac{X_2(s)}{F(s)} = \frac{1}{2} \frac{s^2 + 2s + 7}{s^4 + 3.5s^3 + 12s^2 + 10.5s + 5}$$

Question 22(a)



There are three independent translational displacements.

The mass,

$$(4s^2 + 8s + 5)X_1(s) - 8sX_2(s) - 5X_3(s) = F(s)$$

the tube,

$$-8sX_1(s) + (4s^2 + 16s)X_2(s) - 4sX_3(s) = 0$$

and the point between the spring and damper in series.

$$-5X_1(s) - 4sX_2(s) + (4s + 5)X_3(s) = 0$$

Important: not that since the spring and damper are in series, the point between them can be moved left and right even if the mass and the tube are held in place by some unknown entity. This is why we consider this point's mass balance and you should remember this in future problems and tests.

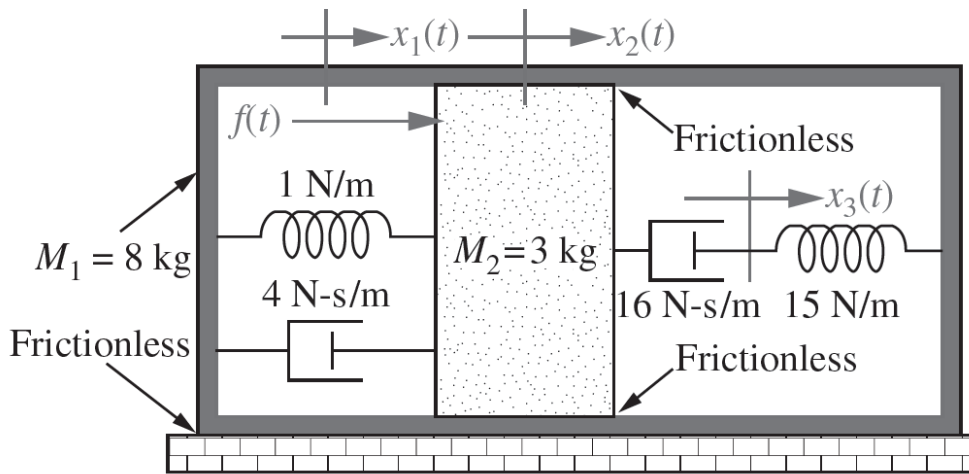
Solving for $X_3(s)$,

$$X_3(s) = \frac{\begin{vmatrix} (4s^2 + 8s + 5) & -8s & F(s) \\ -8s & (4s^2 + 16s) & 0 \\ -5 & -4s & 0 \end{vmatrix}}{\Delta} = \frac{F(s) \begin{vmatrix} -8s & (4s^2 + 16s) \\ -5 & -4s \end{vmatrix}}{\Delta}$$

or,

$$\frac{X_3(s)}{F(s)} = \frac{13s + 20}{4s(4s^3 + 25s^2 + 43s + 15)}$$

Question 22(b)



Very much like in part a of the problem there are three points. Note the extra links between the mass and the tube.

For the Tube

$$(8s^2 + 4s + 16)X_1(s) - (4s + 1)X_2(s) - 15X_3(s) = 0$$

Mass

$$-(4s + 1)X_1(s) + (3s^2 + 20s + 1)X_2(s) - 16sX_3(s) = F(s)$$

and point between damper and spring

$$-15X_1(s) - 16sX_2(s) + (16s + 15)X_3(s) = 0$$

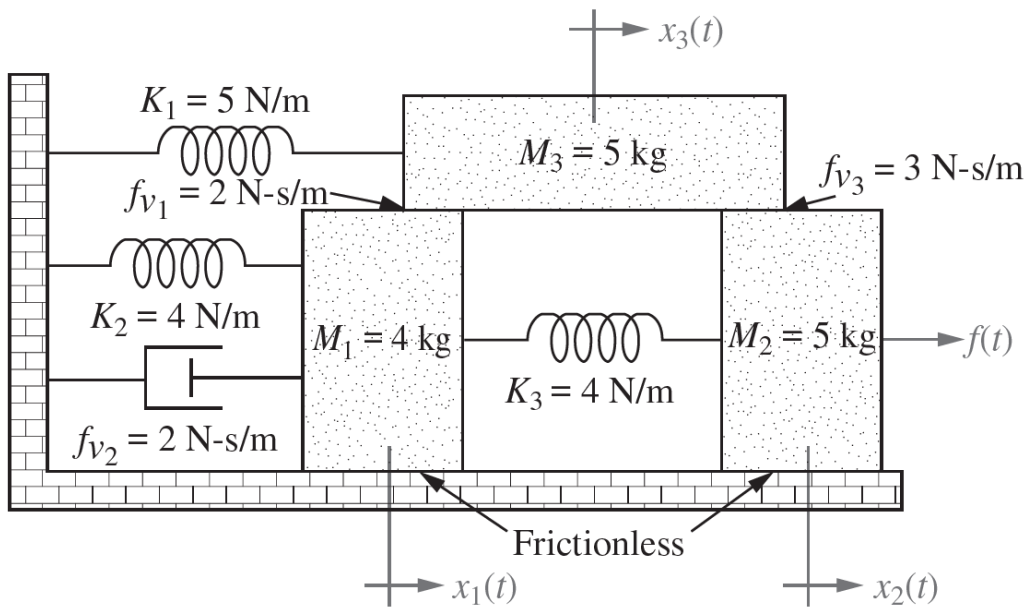
Solving for $X_3(s)$,

$$X_3(s) = \frac{\begin{vmatrix} (8s^2 + 4s + 16) & -(4s+1) & 0 \\ -(4s+1) & (3s^2 + 20s+1) & F(s) \\ -15 & -16s & 0 \end{vmatrix}}{\Delta} = \frac{-F(s) \begin{vmatrix} (8s^2 + 4s + 16) & -(4s+1) \\ -15 & -16s \end{vmatrix}}{\Delta}$$

or

$$\frac{X_3(s)}{F(s)} = \frac{128s^3 + 64s^2 + 316s + 15}{384s^5 + 1064s^4 + 3476s^3 + 165s^2}$$

Question 23



Each mass has a related equation

M_1 :

$$(4s^2 + 4s + 8)X_1(s) - 4X_2(s) - 2sX_3(s) = 0$$

M_2 :

$$-4X_1(s) + (5s^2 + 3s + 4)X_2(s) - 3sX_3(s) = F(s)$$

M_3 :

$$-2sX_1(s) - 3sX_2(s) + (5s^2 + 5s + 5)X_3(s) = 0$$

Note how all the s^2 terms relate to the inertia, s terms are related to the dampers and springs result in constant terms in the laplace domain.